

Waves and Optics

1. Introduction

Waves and optics are the branch of physics that studies wave motion, sound, light, interference, diffraction, polarization, and the physical behavior of wave phenomena in media. It is one of the most important BSc Physics subjects because it connects the mathematics of waves with observable optical and acoustic effects.

At the advanced undergraduate level, the subject usually includes wave equations, superposition, standing waves, acoustics, interference, diffraction, polarization, and interferometry.

2. Wave motion

A wave is a disturbance that transfers energy from one point to another without net transport of matter. Waves may be transverse or longitudinal, and their mathematical description is central to this subject.

Main types

- Transverse waves.
- Longitudinal waves.
- Plane waves.
- Spherical waves.

Basic wave equation

The one-dimensional wave equation is

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$

where v is the wave speed.

Progressive wave

A sinusoidal wave moving in the positive x -direction may be written as

$$y(x, t) = A \sin(kx - \omega t + \phi)$$

with $k = \frac{2\pi}{\lambda}$ and $\omega = 2\pi f$.

3. Velocity of waves

Wave speed depends on the medium and the restoring force. In a string, the speed is determined by tension and linear density.

On a stretched string

$$v = \sqrt{\frac{T}{\mu}}$$

where T is tension and μ is mass per unit length.

In fluids and gases

Sound speed depends on elasticity and density. Newton's formula and Laplace's correction are important historical and conceptual results.

4. Superposition and interference of waves

When two or more waves overlap, the resultant displacement is the algebraic sum of the individual displacements. This principle explains beats, standing waves, and interference.

Constructive interference

Occurs when waves arrive in phase and amplitudes reinforce each other.

Destructive interference

Occurs when waves arrive out of phase and cancel partially or completely.

5. Phase velocity and group velocity

Phase velocity is the speed of a constant phase point, while group velocity is the speed of the envelope or wave packet. These concepts are especially important in dispersive media.

Formulas

$$v_p = \frac{\omega}{k}$$
$$v_g = \frac{d\omega}{dk}$$

In many cases, v_g represents energy or signal transport more meaningfully than v_p .

6. Standing waves

Standing waves are formed by the superposition of two waves of the same frequency and amplitude traveling in opposite directions. They contain nodes and antinodes and are central to vibrations of strings, air columns, and rods.

String fixed at both ends

$$f_n = \frac{n}{2L} v, n = 1, 2, 3, \dots$$

where L is string length.

Open pipe

$$f_n = \frac{n}{2L} v$$

Closed pipe

$$f_n = \frac{(2n-1)}{4L} v$$

These are standard resonance conditions in acoustics.

7. Harmonics and resonance

The fundamental frequency is the lowest natural frequency of a system. Harmonics are integer multiples of the fundamental in ideal systems. Resonance occurs when the forcing frequency matches a natural frequency, producing maximum amplitude.

8. Damped and forced oscillations

In real systems, energy is lost due to friction or resistance, producing damping. Forced oscillations occur when an external periodic force drives the system.

Damped motion

The amplitude decays exponentially with time:

$$A = A_0 e^{-\gamma t}$$

Resonance

At resonance, the amplitude becomes large and the phase shift changes rapidly. This topic is important in mechanical and electrical oscillators.

9. Fourier analysis

Complex waves can be represented as a sum of simple sine and cosine functions. Fourier series is essential for analyzing periodic signals like square waves and sawtooth waves.

General idea

A periodic function can be written as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

This is a major mathematical tool in wave physics.

10. Acoustics

Acoustics deals with sound waves, their production, propagation, loudness, intensity, and interaction with halls and materials.

Important topics

- Audible range of human hearing.
- Intensity and loudness.
- Bel and decibel.
- Reverberation.
- Absorption coefficient.
- Acoustic design of halls.

Decibel scale

$$\beta = 10\log_{10}\left(\frac{I}{I_0}\right)$$

where I_0 is reference intensity.

Reverberation time

Sabine's formula is used to estimate reverberation time in halls:

$$T = \frac{0.161V}{A}$$

where V is room volume and A is total absorption.

11. Ultrasonics

Ultrasonic waves have frequencies above the audible range of humans. They are produced using piezoelectric or magnetostrictive methods and have applications in imaging, testing, cleaning, and detection.

Properties

- High frequency.
- Short wavelength.
- Strong directionality.
- High penetration in certain media.

Applications

- SONAR.
- Medical diagnosis.
- Non-destructive testing.
- Ultrasonic cleaning.
- Measurement of liquid properties.

12. Huygens' principle

Huygens' principle states that every point on a wave-front acts as a source of secondary wavelets, and the new wavefront is the envelope of these wavelets. It is the basis of geometrical and wave optics.

This principle explains reflection, refraction, and propagation of wavefronts in optical media.

13. Coherence

Coherence describes the fixed phase relationship between waves. It is essential for sustained interference.

Types

- Temporal coherence.
- Spatial coherence.

14. Interference of light

Interference occurs when two coherent light waves superpose and produce a redistribution of intensity. It is one of the most important sections of wave optics.

Conditions

- Sources must be coherent.
- Same frequency is required.
- Path difference must remain stable.

Intensity formula

If two waves have amplitudes a_1 and a_2 , then resultant intensity is

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta$$

where δ is phase difference.

15. Young's double slit experiment

Young's double slit experiment is the classic demonstration of interference. It shows bright and dark fringes due to path difference between two coherent sources.

Fringe width

$$\beta = \frac{\lambda D}{d}$$

where D is screen distance and d is slit separation.

16. Fresnel's biprism

Fresnel's biprism creates two virtual coherent sources from a single source. It is used to measure wavelength and demonstrate interference by division of wave-front.

17. Thin film interference

Thin film interference arises due to reflection from the top and bottom surfaces of a thin film. It explains colors in soap bubbles, oil films, and anti-reflection coatings.

Key idea

Interference depends on film thickness, refractive index, wavelength, and phase change on reflection.

18. Newton's rings

Newton's rings are concentric circular fringes produced by interference in the air film between a plano-convex lens and a glass plate. They are used to determine wavelength and radius of curvature.

Ring radius

$$r_n^2 = n\lambda R$$

for dark rings in reflected light, in standard approximations.

19. Michelson interferometer

The Michelson interferometer is a precision instrument used to measure wavelength, refractive index, and small displacements. It works on division of amplitude.

Uses

- Wavelength determination.
- Fringe counting.
- Small path difference measurements.
- Spectral line separation.

20. Diffraction

Diffraction is the bending and spreading of waves around obstacles and apertures. It becomes significant when the size of the aperture is comparable to the wavelength.

Types

- Fresnel diffraction.
- Fraunhofer diffraction.

Single slit diffraction

For a slit of width a , minima occur at

$$a \sin \theta = n\lambda$$

where $n = 1, 2, 3, \dots$

Central maximum

The central maximum is the brightest and widest in single-slit diffraction.

21. Diffraction grating

A diffraction grating has a large number of equally spaced slits, producing sharp maxima and allowing accurate wavelength measurement.

Grating equation

$$d \sin \theta = n\lambda$$

where d is grating element.

Resolving power

Resolving power measures the ability to distinguish close wavelengths.

22. Polarization

Polarization is the phenomenon in which the vibrations of light are restricted to a particular direction. It proves the transverse nature of light.

Methods of producing polarized light

- Reflection.
- Refraction.
- Double refraction.
- Polaroids.
- Scattering.

Malus' law

$$I = I_0 \cos^2 \theta$$

where θ is the angle between transmission axes.

23. Double refraction

In uniaxial crystals, an incident light beam splits into ordinary and extraordinary rays. This is the phenomenon of double refraction.

Crystal types

- Positive crystal.
- Negative crystal.

24. Optical activity

Optically active substances rotate the plane of polarization of plane-polarized light. This is called optical rotation or rotatory polarization.

Specific rotation

$$[\alpha] = \frac{\alpha}{lc}$$

where α is observed rotation, l is path length, and c is concentration.

25. Retarding plates

Quarter-wave and half-wave plates are used to convert between linear, circular, and elliptical polarization.

Functions

- Quarter-wave plate produces circular polarization from suitable linearly polarized light.
- Half-wave plate rotates the plane of polarization.

26. Important formula sheet

- Wave equation:

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$

- Progressive wave:

$$y = A \sin(kx - \omega t + \phi)$$

- String speed:

$$v = \sqrt{\frac{T}{\mu}}$$

- Group velocity:

$$v_g = \frac{d\omega}{dk}$$

- Decibel level:

$$\beta = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

- Sabine's formula:

$$T = \frac{0.161V}{A}$$

- YDSE fringe width:

$$\beta = \frac{\lambda D}{d}$$

- Thin slit minima:

$$a \sin \theta = n\lambda$$

- Grating equation:

$$d \sin \theta = n\lambda$$

- Malus' law:

$$I = I_0 \cos^2 \theta$$

- Specific rotation:

$$[\alpha] = \frac{\alpha}{lc}$$

27. Summary

Waves and optics are a core BSc Physics subject that unifies the study of mechanical waves, sound, and optical wave behavior. It is especially important because it explains interference, diffraction, polarization, and precision instruments used in both experiments and applications.